

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250273054

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Ruymann; Robert

KENO EMPLOYING ONE OR MORE GUARANTEED MATCHING DRAWN SPOTS FOR TOURNAMENTS

Abstract

Gaming systems and methods that provide a play of a keno game having one or more guaranteed matching drawn spots such as where the play of the keno game is part of a play of a tournament, and thereby increases the expected amount won in the play of the keno game for the play of the tournament.

Inventors: Ruymann; Robert (Reno, NV)

Applicant: IGT (Las Vegas, NV)

Family ID: 1000007712013

Appl. No.: 18/585882

Filed: February 23, 2024

Publication Classification

Int. Cl.: G07F17/32 (20060101)

U.S. Cl.:

CPC G07F17/329 (20130101); G07F17/3211 (20130101); G07F17/3267 (20130101);
G07F17/3276 (20130101);

Background/Summary

BACKGROUND

[0001] The present disclosure relates to gaming systems and methods that provide keno employing one or more guaranteed matching drawn spots for tournaments.

[0002] Gaming machines may provide keno games as a primary game or as a secondary game. Gaming machines may provide players awards based on plays of such keno games. Gaming machines may provide tournaments for players.

SUMMARY

[0003] Various embodiments of the present disclosure provide a gaming system including a processor and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to cause a display, by a display device, of a play of a tournament comprising a plurality of plays of a keno game. Each of the plurality of plays of the keno game for the play of the tournament includes: a display, by the display device, of a keno card comprising a plurality of spots; a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card; a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one; and a display, by the display device, of an amount based on the drawn spots that match the player spots. The gaming system also displays an indication of an outcome of the play of the tournament.

[0004] Various other embodiments of the present disclosure provide a gaming system including a processor and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to cause a display, by a display device, of a play of a tournament including a first quantity of plays of a keno game and a second quantity of plays of the keno game, wherein the first quantity is at least one and wherein the second quantity is at least one. Each of the first quantity of plays of the keno game for the play of the tournament includes: a display, by the display device, of a keno card comprising a plurality of spots; a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card; a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one; and a display, by the display device, of an amount based on the drawn spots that match the player spots. Each of the second quantity of plays of the keno game for the play of the tournament includes: a display, by the display device, of the keno card comprising the plurality of spots; a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card; a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the quantity of drawn spots comprise non-guaranteed matching drawn spots; and a display, by the display device, of an amount based on the drawn spots that match the player spots. The gaming system also displays an indication of an outcome of the play of the tournament.

[0005] Various other embodiments of the present disclosure provide a gaming system including a processor and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to cause a display, by a display device, of a play of a tournament comprising a plurality of plays of a keno game. One of the plays of the keno game for the play of the tournament includes: a display, by the display device, of a keno card comprising a plurality of spots; a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card; a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, wherein the display is after shuffling of an order of the guaranteed matching drawn spots for the play of the keno game and the non-guaranteed matching drawn spots

for the play of the keno game, and wherein the display comprises no indications that any of the drawn spots for the play of the keno game are the guaranteed matching drawn spots for the play of the keno game; and a display, by the display device, of an amount based on the drawn spots that match the player spots. The gaming system also displays an indication of an outcome of the play of the tournament.

[0006] Additional features are described in, and will be apparent from, the following Detailed Description and the figures.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING

[0007] FIG. 1 is a flow chart of an example method for operating a gaming system of one example embodiment of the present disclosure that provides a play of a tournament including plays of a keno game, wherein one or more of the plays of the keno game employ one or more guaranteed matching drawn spots.

[0008] FIGS. 2A and 2B are screenshots showing parts of the operation of an example play of a keno game that is part of a play of a tournament displayed by a gaming system and that employs guaranteed matching drawn spots in accordance with one example embodiment of the present disclosure.

[0009] FIG. 3 is a schematic block diagram of one embodiment of an example electronic configuration of an example gaming system disclosed herein.

[0010] FIGS. 4A and 4B are perspective views of example alternative embodiments of the gaming system of further example embodiments of the present disclosure.

[0011] FIG. 4C is a front view of an example personal gaming device of the gaming system of the present disclosure.

DETAILED DESCRIPTION

[0012] In various embodiments, the present disclosure relates to gaming systems and methods of operation thereof that provide one or more plays of a keno game that employ one or more guaranteed matching drawn spots. In various embodiments, the keno game is a game repeatedly played as part of play of a tournament, and one or more plays of the keno game of the play of the tournament employ(s) one or more guaranteed matching drawn spots.

[0013] The following brief background regarding tournaments in casino gaming environments is provided for a better understanding of the present disclosure.

[0014] One known type of tournament includes a bank of gaming machines in a casino at which multiple players simultaneously play (the same game) for a predetermined period of time for the play of the tournament. At the end of the predetermined period of time, and thus the end of the play of the tournament, the player that has won the most credits (or other units) is the winner of that play of the tournament, and is provided with a tournament award. In such tournaments, the players play simultaneously at the gaming machines and are competing against each other.

[0015] Another known type of tournament includes gaming machines such as at one or more casinos that provide the same games that players each play independently for a predetermined period of time as part of the play of the tournament. In these tournaments, each player can select exactly when to play the tournament game during an overall period for the play of the tournament. In such tournaments, the players do not need to play simultaneously during the overall period but are still competing against each other. At the end of the overall period for the play of the tournament, the player that has won the most credits (or other units) is the winner of the play of the tournament, and is provided with a tournament award.

[0016] Various known tournaments includes plays of a reel game where the player simply has to press the play button to cause the reels to spin for the play of the game (without making any

determinations or selections for the play of the game). Such reel games are particularly suited for tournaments because each player can simply press the play button (on the gaming machine they are playing at) as fast as possible to get in as many plays of the reel game as possible during the overall period for the play of the tournament.

[0017] Keno games are often not used for various known tournaments in part because the play of the keno game requires the player to pick numbered spots for the play of the keno game in addition to pressing the play button to cause the play of the keno game. This slows down play and is thus not optimal for timed tournaments such as described above. Even using a pre-pick or an auto-pick feature for play of a keno game or even enabling the player to use the same numbered spots (and thus not requiring have to repeatedly pick the players numbered spots for each play of the keno game), keno games have not been widely employed for tournaments. Keno games are also often not used for various known tournaments in part because the keno game requires 20 drawn spots of the 80 drawn spots, and a displayed payout schedule, wherein if either is enhanced, the players will notice and the plays of the keno game will feel substantially different for regular keno players.

[0018] Various embodiments of the present disclosure provide gaming systems and methods that provide a keno game that can employ zero, one, or more guaranteed matching drawn spots that is particularly suited to be repeatedly played as part of play of a tournament, and thus better enables tournaments that include plays of keno games to be provided to players, and thus particularly suited for players that favor keno games over reel games, and thus would be more likely to participate in tournaments that include plays of a keno game.

[0019] Various embodiments of the present disclosure provide gaming systems and methods that provide a keno game that can employ one or more guaranteed matching drawn spots to increase the average expected payback percentages for one or more the plays of the keno game to make the keno game more suited to be repeatedly played as part of play of a tournament.

[0020] In various embodiments, each play of a keno game of a play of a tournament includes one or more guaranteed matching drawn spots (that increases the average expected payback percentages for those plays of the keno game for the tournament).

[0021] In various embodiments, one or more but not each play of a keno game of a play of a tournament includes one or more guaranteed matching drawn spots (that increases the average expected payback percentages for those plays of the keno game for the tournament).

[0022] The present disclosure contemplates that the quantity of plays of the keno game of a play of a tournament that includes one or more guaranteed matching drawn spots can vary to increase the average expected payback percentage for the plays of the keno game for the play of the tournament in a suitable desired manner, which for a tournament, will provide an enhanced player experience.

[0023] The present disclosure also contemplates that the quantity of guaranteed matching drawn spots in two or more of the plays of the keno game can vary to increase the average expected payback percentage for those plays of the keno game for the play of the tournament in a suitable desired manner.

[0024] In various embodiments, for each play of a keno game of the play of the tournament, the gaming system displays the play of the keno game including one or more keno cards that each includes a plurality of spots arranged on a keno board. In various embodiments, each keno card includes 80 numbered spots arranged in an 8 row and 10 column card configuration, wherein each row includes 10 spots, and each column includes 8 spots. The spots are indicated by respective symbols such as the consecutive numbers 1 to 80. The quantity of keno cards, the quantity of spots on each keno card, the arrangement of the spots on the keno card, the card configuration, and/or the symbols indicating the spots on the keno card can vary in accordance with the present disclosure. In the example embodiments described herein, for brevity, the keno game employs one keno card and card configuration. Such example embodiments are not meant to limit the scope of the present disclosure.

[0025] In various embodiments, for each play of the keno game of the play of the tournament, the

gaming system displays the play of the keno game including a player selection of a quantity of player spots for the keno card (such as two to ten spots from the plurality of spots of the keno card). These player spots of the keno card are each indicated by a player spot indicator (such as but not limited to a specific color or specific symbol). These player spots of the keno card form a player spot set for the play of the keno game. In various embodiment, for each play of a keno game of the play of the tournament, the gaming system enables the player to alternatively auto-pick the player spots or employ player spots selected for a previous play of the keno game (such as to speed of plays of the keno game for the play of the tournament and to make it easier for the players to play the keno game for the play of the tournament).

[0026] In various embodiments, for each play of the keno game of the play of the tournament, the gaming system displays the play of the keno game including a total quantity of drawn spots for the keno card (such as 20 spots of the keno card from the 80 spots on the keno card). These drawn spots are each indicated by a drawn spot indicator (such as but not limited to a keno ball corresponding to that drawn spot). These drawn spots for the keno card form a drawn spot set for the play of the keno game.

[0027] In various embodiments, for each of one or more plays of the keno game of the play of the tournament, the gaming system: (1) selects a first quantity of the drawn spots (such as one or more drawn spots) based on the player selected spots (i.e., the guaranteed matching drawn spots); and (2) randomly selects a second quantity of the drawn spots of the drawn spot set (i.e., the non-guaranteed matching drawn spots). The total quantity of drawn spots for each play of the keno game is the sum of the selected first quantity of the drawn spots and the second quantity of the drawn spots for that play of the keno game. In various embodiments, for each play of the keno game of the play of the tournament wherein the gaming system does not select any drawn spots based on the player selected spots, all of the drawn spots are thus randomly selected.

[0028] The drawn spots that the gaming system selects based on the player spots are sometimes referred to herein as guaranteed matching drawn spots (because each such drawn spot is guaranteed to match one of the player spots for the play of the keno game).

[0029] The drawn spots that the gaming system randomly selects are sometimes referred to herein as non-guaranteed matching drawn spots (because each such drawn spot is not guaranteed to match (but can match) a player spot for the play of the keno game).

[0030] In various embodiments, to guarantee a determined quantity of minimum guaranteed matches between the drawn spots and the player spots for a play of the keno game, after the player picks the player spots for the play of the keno game, the gaming system can select the first quantity (such as one or more) of the drawn spots that each match one of the players spots. After determining the first quantity of minimum guaranteed matches and thus the first quantity of the drawn spots based on the player selected spots, the gaming system can then randomly select a second quantity of the drawn spots. The second quantity is the total quantity of drawn spots for the play of the keno game less the first quantity.

[0031] In various embodiments, for each of one or more plays of the keno game of the tournament, whether or not the gaming system provides a quantity (of one or more) guaranteed matching drawn spots for the play of the keno game is randomly determined.

[0032] In various embodiments, for each of one or more plays of the keno game of the tournament, whether or not the gaming system provides a quantity (of one or more) guaranteed matching drawn spots for the play of the keno game is pre-determined.

[0033] In various embodiments, for each of one or more plays of the keno game of the tournament, whether or not the gaming system provides a quantity (of one or more) guaranteed matching drawn spots for the play of the keno game is based on one or more factors.

[0034] In various embodiments, for each of one or more plays of the keno game of the play of the tournament, the quantity of guaranteed matching drawn spots for the play of the keno game is pre-determined for the play of the keno game.

[0035] In various embodiments, for each of one or more plays of the keno game of the tournament, the quantity of guaranteed matching drawn spot for the play of the keno game is randomly determined for the play of the keno game from a plurality of different quantities.

[0036] In various embodiments, the quantity of guaranteed matching drawn spots can be equal to an amount of matches that are needed for a minimum winning amount in the play of the keno game for the play of the tournament. For example, if 4 player spots out of 10 selected player spots need to be matched for the minimum win in the play of the keno game, the gaming system can select 4 guaranteed matching drawn spots for that play of the game.

[0037] In various embodiments, the quantity of guaranteed matching drawn spots can be less than an amount of matches that are needed for a minimum winning amount in the play of the keno game for the play of the tournament. For example, if 4 player spots out of 7 selected player spots need to be match for the minimum win in the play of the keno game, the gaming system can select 3 guaranteed matching drawn spots for that play of the game. In such example, one additional non-guaranteed drawn spot must be selected for the minimum winning amount in the play of the keno game.

[0038] In various embodiments, the quantity of guaranteed matching drawn spots can be more than an amount of matches that are needed for a minimum winning amount in the play of the keno game for the play of the tournament. For example, if 5 player spots out of 9 selected player spots need to be matched for the minimum win in the play of the keno game, the gaming system can select 6 guaranteed matching drawn spots for that play of the game (and thus more than this minimum).

[0039] In various embodiments, for each of one or more plays of the keno game of the tournament, the quantity of guaranteed matching drawn spots for the play of the keno game is determined from a plurality of different quantities based on one or more characteristics such as but not limited to: (1) a desired average expected payback percentage for that play of the keno game of that play of the tournament; (2) a desired average expected payback percentage for all of the plays of the keno game for that play of the tournament; (3) a desired minimum quantity of matches for that play of the keno game of that play of the tournament; (4) a desired minimum quantity of matches for all of the plays of the keno game for that play of the tournament; and/or (5) a desired win amount for one or more plays of the keno game for that play of the tournament.

[0040] In various embodiments, for each such play of the keno game of the play of the tournament, the gaming system shuffles all of the drawn spots (including the guaranteed matching drawn spots and the non-guaranteed matching drawn spots) to form the drawn spot set. This shuffling step can be performed so that the guaranteed matching drawn spots are not displayed in a group (such as at the beginning or ending of the display of the drawn spots). This prevents the player from knowing which spots are guaranteed matching drawn spots, and/or that the gaming system is even employing guaranteed matching drawn spots for the play of the tournament.

[0041] In various embodiments, the gaming system displays the play of the keno game including: (1) indications of drawn spot set; and (2) indications of matches between the drawn spot set and the player spot set, without displaying any indication that one or more of the drawn spots are guaranteed matching drawn spots. The total quantity of matches will at least be the quantity of guaranteed matching drawn spots, if any.

[0042] Thus, in various embodiments, the gaming system provides improved keno game tournament payback percentages without having to change the payable for the keno game used for the play of the tournament.

[0043] In various embodiments, the gaming system determines any amounts for the play of the keno game based on the total quantity of matches between the drawn spot set and the player spot set, and adds any such determined amounts to a total amount for the player for the play of the tournament. After all the plays of the keno game of the play of the tournament are completed, the gaming system can determine and then display indications of a winner of the play of the tournament based on the relative total amounts accumulated for each player, and determine and

display an indication of the tournament award, as appropriate for the tournament.

[0044] In various embodiments, the gaming system provides improved keno game tournament payback percentages in one or more of the manners described above for each player of the play of the tournament in an equal or an unequal manner.

[0045] In various embodiments, the gaming system provides improved keno game tournament payback percentages in one or more of the manners described above for one or more but not each player of the play of the tournament. In various such embodiments, the gaming system can determine which players receive the improved keno game tournament payback percentages in one or more different suitable manners (such as in a random manner, in a predetermined manner, or in a manner based on one or more player characteristics or factors).

[0046] Thus, in various embodiments, the gaming system provides one or more plays of the keno game with one or more guaranteed matching drawn spots for the play of the tournament for each player of the play of the tournament.

[0047] Thus, in various embodiments, the gaming system provides one or more plays of the keno game with one or more guaranteed matching drawn spots for the play of the tournament for one or more players but not each player of the play of the tournament.

[0048] Turning now to the drawings, FIG. 1 is a flowchart of an example method **100** of operating an example gaming system of the present disclosure. In various embodiments, the example method **100** is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the method **100** is described with reference to the flowchart shown in FIG. 1, other processes of performing the acts associated with this illustrated method may be employed. For example, the order of certain of the illustrated blocks may be changed, certain of the illustrated blocks may be optional, or certain of the illustrated blocks may not be employed.

[0049] This example method described below focuses on the play of the keno game that is one play of a plurality of plays of the keno game that is part of a play of a tournament provided for one player of the tournament. The play of the tournament can include multiple plays of the keno game by the player. The play of the tournament game can additionally include plays of the keno game provided to other players of the tournament. This example method described below thus does not otherwise focus on the method of operation of the play of the tournament which can have any suitable tournament structure such as described above.

[0050] The example method **100** includes, responsive to an occurrence of a triggering event, displaying, via a display device, a play of a keno game that includes a display of a keno card including a plurality of spots, as indicated by block **102**. In various embodiments, the triggering event is an occurrence of a keno game triggering event for the play of the tournament. In various embodiments, the triggering event is an occurrence is completion of a play of a previous keno game in a series of plays of the keno game that are part of the play of the tournament.

[0051] The example method **100** includes receiving, for the keno card, a selection by the player of a plurality of player spots from the plurality of spots of the keno card, as indicated by block **104**. This can be done by the player using an input device to manually select the player spots on the keno card, via an automatic pick mechanism, or in any other suitable manner. In other words, the player spots can be selected by a player or by the gaming system (if the player requests that the gaming system selects the spots). This can also be done by the player maintaining previous player spots from a previous play of the keno game. In various embodiments, the player may select different quantities of spots such as 2 to 10 spots. The payable can take into account the quantity of spots the player selects for a play of the keno game.

[0052] In various embodiments, for a play of a tournament, each player may be required to select a same quantity of spots such as 5 spots or 10 spots of the keno card.

[0053] The method includes displaying, via the display device, for the keno card, indicators of selected player spots selected from the plurality of spots of the keno card, as indicated by block **106**. The indicators of the player spots can be adjacent to the keno card, on the keno card, or both

adjacent to the keno card and on the keno card.

[0054] The example method **100** includes selecting, via a processor, a first quantity guaranteed matching drawn spots, as indicated by block **108**. The gaming system also removes these drawn spots such that they can be subsequently selected by the processor when the processor select the non-guaranteed matching drawn spots. The first quantity can be determined in any suitable manner such as described above.

[0055] The example method **100** includes randomly selecting, via a processor, a second quantity of drawn spots for the play of the keno game, as indicated by block **110**. The amount of this second quantity is based the amount of the first quantity and the total amount of drawn spots that need to be selected for the play of the keno game. For example, if the total amount of drawn spots that need to be selected for the play of the keno game is 20, and the first quantity guaranteed matching drawn spots is 5, then the second quantity of drawn spots (i.e., the non-guaranteed matching drawn spots) for the play of the keno game is 15.

[0056] The example method **100** includes shuffling, via the processor, the order of the selected guaranteed matching drawn spots and the non-guaranteed matching drawn spots, as indicated by block **112**.

[0057] The example method **100** includes displaying, via the display device, indicators of the plurality of all of the drawn spots for the keno card, as indicated by block **114**. The indicators of the drawn spots can be adjacent to the keno card, on the keno card, or both adjacent to the keno card and on the keno card, or both. The indicators of the drawn spots can be balls with numbers thereon or any other suitable indicators. In various embodiments, the indicators do not indicate, for each drawn spot, whether such drawn spot was a guaranteed matching drawn spot or a non-guaranteed matching drawn spot. Thus, the player would not know which spots are guaranteed matching drawn spots and which spots are non-guaranteed matching drawn spots in various embodiments.

[0058] The example method **100** includes displaying, via the display device, indications of matches between the drawn spots and the player spots, as indicated by block **116**. These indications of matches can be displayed on the keno card and/or adjacent to the keno card.

[0059] The example method **100** includes determining and displaying, via the display device, indications of any amounts based on a total quantity of matches between the drawn spots and the player spots, as indicated by block **118**. This determination can also be made based on the quantity of selected player spots. The gaming system also adds these amount to a running total amount for the player for the play of the tournament.

[0060] As indicated above, for each play the keno game for the play of the tournament, each player may accumulate zero or more units (such as but not limited to credits), and at the end of the play of the tournament, the gaming system determines based on the total accumulated units for each player an outcome of the play of the tournament such as one or more the winner(s) of the play of the tournament. The gaming system can also display any tournament awards to provide to any of the winning players of the play of the tournament.

[0061] It should be appreciated that the tournament awards can be any suitable awards such as, but not limited to: (1) monetary credits or currency; (2) non-monetary credits or currency; (3) a modifier (e.g., a multiplier) used to modify one or more awards; (4) one or more free plays of a game; (5) one or more plays of a bonus game (e.g., a free spin of an award wheel, a free spin of the award generator, etc.); (6) one or more lottery based awards (e.g., one or more lottery or drawing tickets); (7) a wager match for one or more plays of the wagering game; (8) an increase in an average expected payback percentage of a bonus game and/or an average expected payback percentage of a primary game for one or more plays; (9) one or more comps (such as a free meal or a free night's stay at a hotel); (10) one or more bonus or promotional credits usable for online play; (11) one or more player tracking points; (12) a multiplier for player tracking points; (13) an increase in a membership or player tracking level; (14) one or more coupons or promotions usable within a gaming establishment or outside of the gaming establishment (e.g., a 20% off coupon for

use at a retail store or a promotional code providing a deposit match for use at an online casino); (15) an access code usable to unlock content on the Internet; (16) a progressive award; (17) a high value product or service (such as a car); and/or (18) a low value product or service.

[0062] FIGS. 2A and 2B show certain example screen shots displayed by a display device of a gaming system displaying parts of an example play of a keno game that is part of a play of a tournament and that employs guaranteed matching drawn spots for the play of the keno game in accordance with one example embodiment of the present disclosure.

[0063] In the example shown in FIGS. 2A and 2B, the gaming system displays, via a display device **200**, a play of the keno game including a keno card **210** having a set of spots respectively represented by the numbers 1 through 80. It should be appreciated that any other suitable configuration of keno cards, spots, symbols representing the spots, and various different indications, and any other suitable different quantity of keno cards can be utilized in accordance with the present disclosure. In this example, the spots are associated with the numbers 1 through 80. It should be appreciated that the spots can be associated with other numbers or other symbols in accordance with the present disclosure.

[0064] In this example embodiment, the gaming system also displays via the display device **200**, a plurality of meters including: (i) a win amount meter **232** that displays any amounts won for the each play of the keno game (in credits or in another form); (ii) a bet meter **234** that displays any bets placed on each play of the keno game (in credits or in another form); and (iii) a credit meter **236** that displays the credit balance of the player (in credits or another form). The gaming system can display other indicators and information such as but not limited to a paytable for the play(s) of the keno game that displays amounts that can be won in each play of the keno game; and information regarding the features of the keno game. In this example embodiment, the gaming system also displays, via the display device, a plurality of input buttons actuatable by the player including: (i) a start button **252** that, when actuated by the player, causes the gaming system to initiate a play of the keno game; (ii) a clear button **254** that, when actuated by the player, causes the gaming system to remove any keno spots from the player spot set; (iii) a spot pick button **256** that, when actuated by the player, causes the gaming system to enable the player to pick the spots (represented by numbers) to include in the player spot set for the keno card; (iv) a quick pick button **258** that, when actuated by the player, causes the gaming system to randomly determine the spots to include in the player spot set for the keno card; and (v) increase and decrease bet buttons (not shown) that, when actuated by the player, respectively increase and decrease the amount to be wagered (such as the quantity of credits or currency to be wagered) as displayed by the bet meter. The gaming system enables the player to select which of the keno spots to include in the player spot set for the play of the keno game. These configuration of these input buttons and display meters can vary in accordance with the present disclosure.

[0065] In the example play of the keno game, responsive to receiving the actuation of the quick pick button **258**, the gaming system selects and displays a set of player spots for the player spot set for the keno card **210** for the play of the keno game. In this example embodiment, the player spots are represented by the numbers 6, 21, 36, 46, 51, 73, and 76 from the plurality of keno spots (not individually labeled) of the keno card **210**. These player spots are indicated by the square box indicators respectively surrounding such numbers on the keno card **210** in FIG. 2A. FIG. 2A thus shows that the gaming system employs the 7 keno spots represented by numbers including 6, 21, 36, 46, 51, 73, and 76 from the plurality of 80 keno spots of the keno card **210** that the gaming system selected for the player in the example play of the keno game.

[0066] At this point in time, without any indication to the player, the gaming system determines how many, if any, guaranteed matching drawn spots the gaming system should select for this play of the keno game. The determination of how many guaranteed matching drawn spots can be determined in any suitable manner such based on the desired winning percentage for the play of the keno game or the desired minimum quantity of matches desired for this play of the keno game.

[0067] In this example, the gaming system determines to select four guaranteed matching drawn spots for the play of the keno game. In this example, the gaming system selects, without any indication to the player, the spots 6, 21, 36, and 46 to be the guaranteed matching drawn spots for the play of the keno game. These selected guaranteed matching drawn spots are indicated in phantom in FIG. 2B (but would not be displayed to the player in this manner in various embodiments). In other embodiments, the gaming system can display one or more of the selected guaranteed matching drawing spots to the player. The gaming system also removes these selected guaranteed drawn spots from the pool of selectable spots for this play of the keno game such that they can not be subsequently selected by the processor when the processor randomly selects the non-guaranteed matching drawn spots for this play of the keno game.

[0068] The gaming system determines, without any indication to the player, how many, if any, non-guaranteed matching drawn spots the gaming system should select for the play of the keno game. This determination is based on the total quantity of drawn spots for the play of the game less the quantity of guaranteed matching drawn spots selected for the play of the keno game. In this example, the quantity of non-guaranteed matching drawn spots for the play of the keno game is 16 (which is 20 total drawn spots minus the 4 guaranteed matching drawn spots that were selected). The gaming system then randomly selects, without any indication to the player, the 16 non-guaranteed matching drawn spots for the play of the keno game that in this example are: 3, 11, 16, 20, 25, 28, 29, 35, 47, 49, 51, 53, 56, 68, 73, and 75.

[0069] The gaming system then shuffles, without any indication to the player, the order of the selected non-guaranteed matching drawn spots 6, 21, 36, and 46 and the selected guaranteed matching drawn spots 3, 11, 16, 20, 25, 28, 29, 35, 47, 49, 51, 53, 56, 68, 73, and 75 to determine a display order. The gaming system then displays the drawn spots in the display order that in this case is from top to bottom and specifically: 29, 16, 36, 56, 20, 53, 68, 21, 28, 6, 46, 35, 11, 51, 47, 73, 3, 25, 49, and 75 as shown in FIG. 2B.

[0070] FIG. 2B thus shows a screenshot at a subsequent point in time in the play of the example keno game at which the gaming system has displayed the drawn spots that form the drawn spot set including the keno spots represented by the spots in the shuffled order 29, 16, 36, 56, 20, 53, 68, 21, 28, 6, 46, 35, 11, 51, 47, 73, 3, 25, 49, and 75. It should be appreciated that the order of display of these drawn spots do not indicate to the player which of the drawn spots are guaranteed matching drawn spots. Thus, in this example embodiment, the player cannot tell which of these drawn spots are the guaranteed matching drawn spots and which are the non-guaranteed matching drawn spots. In fact, in this example embodiment, the player does not know that there are any guaranteed matching drawn spots at all.

[0071] FIG. 2B also shows that the gaming system has also indicated which player spots have been matched by the drawn spots by the HIT indicators adjacent to the keno card **110**. These indications can be any suitable indications in accordance with the present disclosure. At this point in the play of the keno game, the drawn spots have matched (e.g., Hit) all seven of the player spots 6, 21, 36, 46, 51, 73 and 76. The gaming system also indicates that the total amount of 5000 credits won for the seven matches. In this example, with the guaranteed matching drawn spots thus provided at least a 4 match win and increased the probability of more than a 4 match win.

[0072] This configuration increase the amounts of the units such as credits accumulated for this play of the keno game of the play of the tournament game, and thus increases the average expected payback percentage without changing the paytable of the keno game for the play of the tournament.

[0073] It should be appreciated from that above that various embodiments of the present disclosure provide a gaming system configured to: cause a display, by a display device, of a play of a tournament comprising a plurality of plays of a keno game, wherein each of the plurality of plays of the keno game for the play of the tournament comprises: (a) a display, by the display device, of a keno card comprising a plurality of spots, (b) a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, (c) a display, by

the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, and (d) a display, by the display device, of an amount based on the drawn spots that match the player spots; and (2) cause a display, by a display device, of an indication of an outcome of the play of the tournament. In various such embodiments, the gaming system is configured to, for each of the plurality of plays of the keno game for the play of the tournament, cause the display, by the display device, of the drawn spots for that play of the keno game after shuffling an order of the guaranteed matching drawn spots for that play of the keno game and the non-guaranteed matching drawn spots for that play of the keno game. In various such embodiments, the plurality of plays of the keno game for the play of the tournament are all of the plays of the keno game for the play of the tournament. In various such embodiments, the plurality of plays of the keno game for the play of the tournament are less than all of the plays of the keno game for the play of the tournament, and for at least one additional play of the keno game for the play of the tournament, and the gaming system is configured to, for each additional play of the keno game for the play of the tournament, cause: a display, by the display device, of the keno card comprising the plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the plurality of drawn spots comprise non-guaranteed matching drawn spots, and a display, by the display device, of an amount based on the drawn spots that match the player spots. In various such embodiments, the gaming system is configured to, for each of the plurality of plays of the keno game for the play of the tournament, cause the display, by the display device, of no indications that any of the drawn spots for that play of the keno game are the guaranteed matching drawn spots for that play of the keno game. In various such embodiments, the gaming system is configured to, for one of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for one of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for each of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for each of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

[0074] It should further be appreciated from that above that various embodiments of the present disclosure provide a gaming system configured to: cause a display, by a display device, of a play of a tournament comprising a first quantity of plays of a keno game and a second quantity of plays of the keno game, wherein the first quantity is at least one, wherein the second quantity is at least one, and wherein each of the first quantity of plays of the keno game for the play of the tournament comprises: (a) a display, by the display device, of a keno card comprising a plurality of spots, (b) a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, (c) a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity

of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, and (d) a display, by the display device, of an amount based on the drawn spots that match the player spots, and wherein each of the second quantity of plays of the keno game for the play of the tournament comprises: (i) a display, by the display device, of the keno card comprising the plurality of spots, (ii) a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, (iii) a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the quantity of drawn spots comprise non-guaranteed matching drawn spots, and (iv) a display, by the display device, of an amount based on the drawn spots that match the player spots; and cause a display, by a display device, of an indication of an outcome of the play of the tournament. In various such embodiments, the gaming system is configured to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the display, by the display device, of the drawn spots for that play of the keno game after shuffling an order of the guaranteed matching drawn spots for that play of the keno game and the non-guaranteed matching drawn spots for that play of the keno game. In various such embodiments, the gaming system is configured to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the display, by the display device, of no indications that any of the drawn spots for that play of the keno game are the guaranteed matching drawn spots for that play of the keno game. In various such embodiments, the gaming system is configured to, for one of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for one of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game. In various such embodiments, the gaming system is configured to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

[0075] It should further be appreciated from that above that various embodiments of the present disclosure provide a gaming system configured to: (1) cause a display, by a display device, of a play of a tournament comprising a plurality of plays of a keno game, wherein one of the plays of the keno game for the play of the tournament comprises: (a) a display, by the display device, of a keno card comprising a plurality of spots, (b) a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, (c) a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, wherein the display is after shuffling of an order of the guaranteed matching drawn spots for the play of the keno game and the non-guaranteed matching drawn spots for the play of the keno game, and wherein the display comprises no indications that any of the drawn spots for the play of the keno game are the guaranteed matching drawn spots for the play of the keno game, and (d) a display, by the display device, of an amount based on the drawn spots that match the player spots; and (2) cause a display,

by a display device, of an indication of an outcome of the play of the tournament. In various such embodiments, the gaming system is configured to, for an additional one of the plays of the keno game for the play of the tournament, cause: a display, by the display device, of the keno card comprising the plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the quantity of drawn spots comprise non-guaranteed matching drawn spots, and a display, by the display device, of an amount based on the drawn spots that match the player spots. In various such embodiments, the gaming system is configured to, for the play of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for the play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for the play of the keno game. In various such embodiments, the gaming system is configured to, for the play of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for the play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for the play of the keno game.

[0076] It should also be appreciated that the present disclosure provides an improvement in gaming technology by providing one or more of: (1) a new feature for a keno game that can dramatically increase the awards that can be won by the player and increase volatility; (2) guaranteed matching drawn spots; and (3) a new display for a keno game. Such features increase player time on and use of the gaming system of the present disclosure, and thus decrease wear and tear on other gaming systems in a casino.

[0077] It should also be appreciated that in different embodiments, one or more of any determination disclosed herein is/are predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a generated symbol or symbol combination, determined independent of a generated symbol or symbol combination, determined based on a random determination by the central controller, determined independent of a random determination by the central controller, determined based on a random determination at the gaming system, determined independent of a random determination at the gaming system, determined based on at least one play of at least one game, determined independent of at least one play of at least one game, determined based on a player's selection, determined independent of a player's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the player's primary game wager, determined independent of the player's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the player (i.e., a player tracking status), determined independent of a status of the player (i.e., a player tracking status), determined based on one or more other determinations disclosed herein, determined independent of any other determination disclosed herein or determined based on any other suitable method or criteria.

Gaming Systems

[0078] The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

[0079] The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A “gaming system” as used herein refers to various configurations of: (a) one or more servers; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices. Thus, in various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more servers; (b) one or more

personal gaming devices in combination with one or more servers; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more servers in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single server; and/or (j) a plurality of servers in combination with one another. For brevity and clarity and unless specifically stated otherwise, “EGM” as used herein represents one EGM or a plurality of EGMs, “personal gaming device” as used herein represents one personal gaming device or a plurality of personal gaming devices, and “server” as used herein represents one server or a plurality of servers.

[0080] As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in combination with a server. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the server through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a server through a data network.

[0081] In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a server, the server is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the server. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the server is configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the server and the EGM (or personal gaming device). The at least one processor of the server is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the server. One, more than one, or each of the functions of the server may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the server.

[0082] In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the server. In such “thin client” embodiments, the server remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such “thick client” embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

[0083] In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices)

are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a server, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the server in a thin client configuration.

[0084] In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the server. In one example, the EGMs (or personal gaming devices) and the server are located in a gaming establishment or a portion of a gaming establishment.

[0085] In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the server. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the server is located; or (b) in a gaming establishment different from the gaming establishment in which the server is located. In another example, the server is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a server and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN, though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

[0086] In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the server identifies a player before enabling that player to place any wagers on any plays of any wagering games. In one example, the server identifies the player by requiring a player account of the player to be logged into via an input of a unique username and password combination assigned to the player. The server may, however, identify the player in any other suitable manner, such as by validating a player tracking identification number associated with the player; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique player identification number associated with the player by the server; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the server identifies the player, the server enables

placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games and displays those plays via the Internet browser of the EGM (or personal gaming device).

[0087] The server and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for players to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with players.

[0088] FIG. 3 is a block diagram of an example EGM **1000** and FIGS. 4A and 4B include two different example EGMs **2000a** and **2000b**. The EGMs **1000**, **2000a**, and **2000b** are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs **1000**, **2000a**, and **2000b**. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device **2000c** of FIG. 4C) may include some or all of the below components.

[0089] In these embodiments, the EGM **1000** includes a master gaming controller **1012** configured to communicate with and to operate with a plurality of peripheral devices **1022**.

[0090] The master gaming controller **1012** includes at least one processor **1010**. The at least one processor **1010** is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface **1006** of the master gaming controller **1012**; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices **1022** (such as input/output devices); and/or (5) controlling the peripheral devices **1022**. In certain embodiments, one or more components of the master gaming controller **1012** (such as the at least one processor **1010**) reside within a housing of the EGM (described below), while in other embodiments at least one component of the master gaming controller **1012** resides outside of the housing of the EGM.

[0091] The master gaming controller **1012** also includes at least one memory device **1016**, which includes: (1) volatile memory (e.g., RAM **1009**, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other suitable forms); (2) non-volatile memory **1019** (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs **1008**); (4) read-only memory; and/or (5) a secondary memory storage device **1015**, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM of the present disclosure. In certain embodiments, the at least one memory device **1016** resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device **1016** resides outside of the housing of the EGM. In these embodiments, any combination of

one or more computer readable media may be utilized. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random-access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0092] A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

[0093] The at least one memory device **1016** is configured to store, for example: (1) configuration software **1014**, such as all the parameters and settings for a game playable on the EGM; (2) associations **1018** between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable the at least one processor **1010** to communicate with the peripheral devices **1022**; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hiperlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller **1012** communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as peripherals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller **1012** include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

[0094] As will be appreciated by one skilled in the art, aspects of the present disclosure may be illustrated and described herein in any of a number of patentable classes or context including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, aspects of the present disclosure may be implemented entirely hardware, entirely software (including firmware, resident software, micro-code, etc.) or combining software and hardware implementation that may all generally be referred to herein as a “circuit,” “module,” “component,” or “system.” Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable media having computer readable program code embodied thereon.

[0095] Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the “C” programming language, Visual Basic, Fortran 2003, Perl, COBOL 2002, PHP, ABAP, dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or

entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (Saas).

[0096] Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0097] These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0098] In certain embodiments, the at least one memory device **1016** is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device **1016** of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, payable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In certain such embodiments, an operator (such as a gaming establishment operator) and/or a player uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

[0099] The at least one memory device **1016** also stores a plurality of device drivers **1042**.

Examples of different types of device drivers include device drivers for EGM components and device drivers for the peripheral components **1022**. Typically, the device drivers **1042** utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet **175**, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the

at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different type of card reader when device drivers for both card readers are stored in the at least one memory device.

[0100] In certain embodiments, the software units stored in the at least one memory device **1016** can be upgraded as needed. For instance, when the at least one memory device **1016** is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device **1016** from the master game controller **1012** or from some other external device. As another example, when the at least one memory device **1016** includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device **1016** can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another example, when the at least one memory device **1016** uses flash memory **1019** or EPROM **1008** units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

[0101] In some embodiments, the at least one memory device **1016** also stores authentication and/or validation components **1044** configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, user input device components, information received from one or more user input devices, information stored in the at least one memory device **1016**, etc.

[0102] In certain embodiments, the peripheral devices **1022** include several device interfaces, such as: (1) at least one output device **1020** including at least one display device **1035**; (2) at least one input device **1030** (which may include contact and/or non-contact interfaces); (3) at least one transponder **1054**; (4) at least one wireless communication component **1056**; (5) at least one wired/wireless power distribution component **1058**; (6) at least one sensor **1060**; (7) at least one data preservation component **1062**; (8) at least one motion/gesture analysis and interpretation component **1064**; (9) at least one motion detection component **1066**; (10) at least one portable power source **1068**; (11) at least one geolocation module **1076**; (12) at least one user identification module **1077**; (13) at least one player/device tracking module **1078**; and (14) at least one information filtering module **1079**.

[0103] The at least one output device **1020** includes at least one display device **1035** configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a player's player tracking status (as described below); (c) a secondary or upper display device in addition to the central display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash, account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM **2000a** illustrated in FIG. 4A includes a central display device **2116**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**. The example EGM **2000b** illustrated in FIG. 4B includes a central display device **2116**, an upper display device **2118**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**.

[0104] In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a

display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touchscreen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

[0105] The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

[0106] In various embodiments, the at least one output device **1020** includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the player. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a ticket printer and dispenser **2136**.

[0107] In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the player following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the player in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the player; via a transfer of funds onto an electronically recordable identification card or smart card of the player; or via sending a virtual ticket having a monetary value to an electronic device of the player.

[0108] While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

[0109] In certain embodiments, the at least one output device **1020** is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs **2000a** and **2000b** illustrated in FIGS. 4A and 4B each include a plurality of speakers **2150**. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract players to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential players to the EGM. The videos may be customized to provide any appropriate information.

[0110] The at least one input device **1030** may include any suitable device that enables an input signal to be produced and received by the at least one processor **1010** of the EGM.

[0111] In one embodiment, the at least one input device **1030** includes a payment device configured

to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a player identification card reader into which a player identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a combined bill and ticket acceptor **2128** and a coin slot **2126**.

[0112] In one embodiment, the at least one input device **1030** includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a player, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that player to fund the EGM. When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

[0113] In certain embodiments, the at least one input device **1030** includes at least one wagering or betting device. In various embodiments, the one or more wagering or betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

[0114] In various embodiments, the at least one input device **1030** includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a player appropriately funds the EGM and places a wager, the EGM activates the game play activation device to enable the player to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a game play activation device in the form of a game play initiation button **2132**. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

[0115] In other embodiments, the at least one input device **1030** includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the

EGM receives an actuation of the cashout device from a player and the player has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the player's credit balance. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a cashout device in the form of a cashout button **2134**.

[0116] In various embodiments, the at least one input device **1030** includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a plurality of such buttons **2130**.

[0117] In certain embodiments, the at least one input device **1030** includes a touchscreen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touchscreen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

[0118] In embodiments including a player tracking system, as further described below, the at least one input device **1030** includes a card reader in communication with the at least one processor of the EGM. The example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B** each include a card reader **2138**. The card reader is configured to read a player identification card inserted into the card reader.

[0119] The at least one wireless communication component **1056** includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component **1056** transmits electrical, electromagnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

[0120] The at least one wired/wireless power distribution component **1058** includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component **1058** includes a magnetic induction system that is configured to provide wireless power to one or more user input devices near the EGM. In one embodiment, a user input device docking region is provided, and includes a power distribution component that is configured to recharge a user input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component **1058** is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

[0121] In certain embodiments, the at least one sensor **1060** includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor **1060** may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a predetermined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., players, casino employees, etc.), devices (e.g., user input devices), and/or systems within a predetermined proximity to the EGM.

[0122] The at least one data preservation component **1062** is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation system **1062** may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

[0123] The at least one motion/gesture analysis and interpretation component **1064** is configured to analyze and/or interpret information relating to detected player movements and/or gestures to determine appropriate player input information relating to the detected player movements and/or gestures. For example, in one embodiment, the at least one motion/gesture analysis and interpretation component **1064** is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a player; interpret the player's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the player; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

[0124] The at least one portable power source **1068** enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM **300** includes one or more rechargeable batteries.

[0125] The at least one geolocation module **1076** is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module **1076** is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module **1076** is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

[0126] The at least one user identification module **1077** is configured to determine the identity of the current user or current owner of the EGM. For example, in one embodiment, the current user is required to perform a login process at the EGM in order to access one or more features.

Alternatively, the EGM is configured to automatically determine the identity of the current user based on one or more external signals, such as an RFID tag or badge worn by the current user and that provides a wireless signal to the EGM that is used to determine the identity of the current user. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized users from accessing confidential or sensitive information.

[0127] The at least one information filtering module **1079** is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays **1035** of the EGM.

[0128] In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, solenoids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices.

[0129] As generally described above, in certain embodiments, such as the example EGMs **2000a** and **2000b** illustrated in FIGS. **4A** and **4B**, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM.

Further, the EGM is configured such that a player may operate it while standing or sitting. In various embodiments, the EGM is positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a player may operate typically while sitting. As illustrated by the different example EGMs **2000a** and **2000b** shown in FIGS. **4A** and **4B**, EGMs may have varying housing and display configurations.

[0130] In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained

approval from a regulatory gaming commission.

[0131] The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

[0132] In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as “primary games”) and/or any secondary or bonus games or other functions (referred to herein as “secondary games”) displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a player; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a player.

[0133] As generally explained above, in various embodiments in which the gaming system includes a server and a changeable EGM, the at least one memory device of the server stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one changeable EGM as a secondary game to be played simultaneously with a play of a primary game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

[0134] In operation of such embodiments, the server is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the server to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

[0135] In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated

probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

[0136] In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award.

[0137] In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided, the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards.

[0138] In certain embodiments in which the gaming system includes a server and an EGM, the EGM is configured to communicate with the server for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the server monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the server. In this embodiment, the accounting and gaming information system includes: (a) a player database configured to store player profiles, (b) a player tracking module configured to track players (as described below), and (c) a credit system configured to provide automated transactions.

[0139] As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) can include any suitable games and/or wagering games, such as, but not limited to electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

[0140] In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system.

[0141] In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs,

the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award.

[0142] As generally noted above, in addition to providing winning credits or other awards for one or more plays of the primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of player excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

[0143] In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a “BONUS” symbol appearing on three adjacent reels along a payline following a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

[0144] In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

[0145] In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, which is obtained, a given number of secondary game wagering points or credits is accumulated in a “secondary game meter” configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

[0146] In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments' entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple “buy-in.” For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager “buys-in” to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game, or a wager of a

designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

[0147] In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable players of those EGMs to work in conjunction with one another, such as by enabling the players to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable players of those EGMs to compete against one another for one or more awards. In one such embodiment, the EGMs enable the players of those EGMs to participate in one or more gaming tournaments for one or more awards.

[0148] In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a player's gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a player is issued a player identification card that has an encoded player identification number that uniquely identifies the player. When the player's playing tracking card is inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the player identification number off the player tracking card to identify the player. The gaming system timely tracks any suitable information or data relating to the identified player's gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

[0149] In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different embodiments, for one or more players, the player tracking system includes the player's account number, the player's card number, the player's first name, the player's surname, the player's preferred name, the player's player tracking ranking, any promotion status associated with the player's player tracking card, the player's address, the player's birthday, the player's anniversary, the player's recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device.

[0150] In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the player must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an “app”) installed on the personal gaming device before the player can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices (such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers,

the one or more servers make various determinations based on the inputs and determine content to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

[0151] In certain such embodiments, the one or more servers must identify the player before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the player must identify herself to the one or more servers, such as by inputting the player's unique username and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

[0152] Once identified, the one or more servers enable the player to establish an account balance from which the player can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the player to initiate an electronic funds transfer to transfer funds from a bank account to the player's account balance. In other embodiments, the one or more servers enable the player to make a payment using the player's credit card, debit card, or other suitable device to add money to the player's account balance. In other embodiments, the one or more servers enable the player to add money to the player's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the player to cash out the player's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the player, or by initiating printing of a voucher at a kiosk in a gaming establishment.

[0153] In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out players' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the player's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

[0154] If the payment server determines that the player's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the player that the player's account balance is too low to place the desired wager. If the payment server determines that the player's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server.

[0155] The game server then determines an outcome and any associated award for the play of the game. The game server notifies the payment server of any nonzero award, and the payment server increases the player's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

[0156] In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more

servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area. [0157] In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device.

Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the player to play games on the EGM remotely via the personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area.

[0158] In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a player's gaming experience with the player's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the player's wall, newsfeed, or similar area of the social networking website accessible by the player's connections (and in certain cases the public) such that the player's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the player's likes or dislikes or the player's list of connections. In certain embodiments, the gaming system enables the player to link the player's player account to the player's social networking account(s). This enables the gaming system to, once it identifies the player and initiates a gaming session (such as via the player logging in to a website (or an application) on the player's personal gaming device or via the player inserting the player's player tracking card into an EGM), link that gaming session to the player's social networking account(s). In other embodiments, the gaming system enables the player to link the player's social networking account(s) to individual gaming sessions when desired by providing the required login information.

[0159] For instance, in one embodiment, if a player wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to play). In another embodiment, if a player joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network sever to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see (and to entice them to fill the vacancy). In another embodiment, if the player consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the player's wall (or other suitable area) of the social networking website for the player's connections to see. In another embodiment, the gaming system enables the player to recommend a game to the player's connections by posting a recommendation to the player's wall (or other suitable area) of the social networking website.

[0160] Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

[0161] For instance, EGMs are highly regulated to ensure fairness, and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are

implemented in EGMs that differ significantly from those of general-purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

[0162] At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general-purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general-purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general-purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

[0163] Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general-purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored, or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the player, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the player. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general-purpose computing device.

[0164] A second difference between EGMs and general-purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a player of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

[0165] A third difference between EGMs and general-purpose computing devices is authentication-EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code

using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code.

[0166] A fourth difference between EGMs and general-purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general-purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

[0167] To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software components and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

[0168] Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to “re-trigger” the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In other words, the watchdog timer always functions from the time power is applied to the board.

[0169] Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general-purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

[0170] As described above, certain EGMs are state-based machines. Different functions of the game provided by the EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the player's wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state

until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

[0171] Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or all do not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

[0172] To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write capability which meets or exceeds minimum read/write access characteristics (such as at least 5.08 Mbytes/sec (Read) and/or at least 38.0 Mbytes/sec (Write)). Memory devices that meet or exceed the above criteria may be referred to as “fault-tolerant” memory devices.

[0173] Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly, battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

[0174] Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

[0175] As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume, and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a player is required to make a number of selections on a video display screen. When a malfunction has occurred after the player has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the player. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

[0176] Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of the graphical presentation that was previously presented on the EGM and the state of the EGM

(e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a player may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the player was correct or not in the player's assertion.

[0177] Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that differ from the “standard” EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

[0178] The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

[0179] Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

[0180] Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

[0181] Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The purpose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third-party verification device. Once the trusted memory device is verified as authentic and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives.

[0182] In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., “unalterable memory”) such as EPROMS, PROMS, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

[0183] According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the

identity of the trusted information source. For example, the trusted information source and the remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities. [0184] EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information might be designed to detect tampering attempts and clear or erase itself when an attempt at tampering has been detected.

[0185] Mass storage devices used in a general-purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present.

[0186] It should be appreciated that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. For example, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. In another example, the terms “including” and “comprising” and variations thereof, when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. Additionally, a listing of items does not imply that any or all of the items are mutually exclusive nor does a listing of items imply that any or all of the items are collectively exhaustive of anything or in a particular order, unless expressly specified otherwise. Moreover, as used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It should be further appreciated that headings of sections provided in this document and the title are for convenience only and are not to be taken as limiting the disclosure in any way. Furthermore, unless expressly specified otherwise, devices that are in communication with each other need not be in continuous communication with each other and may communicate directly or indirectly through one or more intermediaries.

[0187] Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. For example, a description of an embodiment with several components in communication with each other does not imply that all such components are required, or that each of the disclosed components must communicate with every other component. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present disclosure. As such, these changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended technical scope. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A gaming system comprising: a processor; and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a

display device, of a play of a tournament comprising a plurality of plays of a keno game, wherein each of the plurality of plays of the keno game for the play of the tournament comprises: a display, by the display device, of a keno card comprising a plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, and a display, by the display device, of an amount based on the drawn spots that match the player spots; and cause a display, by a display device, of an indication of an outcome of the play of the tournament.

2. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the plurality of plays of the keno game for the play of the tournament, cause the display, by the display device, of the drawn spots for that play of the keno game after shuffling an order of the guaranteed matching drawn spots for that play of the keno game and the non-guaranteed matching drawn spots for that play of the keno game.

3. The gaming system of claim 1, wherein the plurality of plays of the keno game for the play of the tournament are all of the plays of the keno game for the play of the tournament.

4. The gaming system of claim 1, wherein the plurality of plays of the keno game for the play of the tournament are less than all of the plays of the keno game for the play of the tournament, and for at least one additional play of the keno game for the play of the tournament, the plurality of instructions, when executed by the processor, cause the processor to, for each additional play of the keno game for the play of the tournament, cause: a display, by the display device, of the keno card comprising the plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the plurality of drawn spots comprise non-guaranteed matching drawn spots, and a display, by the display device, of an amount based on the drawn spots that match the player spots.

5. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the plurality of plays of the keno game for the play of the tournament, cause the display, by the display device, of no indications that any of the drawn spots for that play of the keno game are the guaranteed matching drawn spots for that play of the keno game.

6. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to, for one of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

7. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to, for one of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

8. The gaming system of claim 1, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

9. The gaming system of claim 1, wherein the plurality of instructions, when executed by the

processor, cause the processor to, for each of the plurality of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

10. A gaming system comprising: a processor; and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of a play of a tournament comprising a first quantity of plays of a keno game and a second quantity of plays of the keno game, wherein the first quantity is at least one, wherein the second quantity is at least one, wherein each of the first quantity of plays of the keno game for the play of the tournament comprises: a display, by the display device, of a keno card comprising a plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, and a display, by the display device, of an amount based on the drawn spots that match the player spots, and wherein each of the second quantity of plays of the keno game for the play of the tournament comprises: a display, by the display device, of the keno card comprising the plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the quantity of drawn spots comprise non-guaranteed matching drawn spots, and a display, by the display device, of an amount based on the drawn spots that match the player spots; and cause a display, by a display device, of an indication of an outcome of the play of the tournament.

11. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the display, by the display device, of the drawn spots for that play of the keno game after shuffling an order of the guaranteed matching drawn spots for that play of the keno game and the non-guaranteed matching drawn spots for that play of the keno game.

12. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the display, by the display device, of no indications that any of the drawn spots for that play of the keno game are the guaranteed matching drawn spots for that play of the keno game.

13. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for one of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

14. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for one of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

15. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

16. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to, for each of the first quantity of plays of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for that play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for that play of the keno game.

17. A gaming system comprising: a processor; and a memory device storing a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of a play of a tournament comprising a plurality of plays of a keno game, wherein one of the plays of the keno game for the play of the tournament comprises: a display, by the display device, of a keno card comprising a plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein the quantity of drawn spots comprise a first quantity of guaranteed matching drawn spots and a second quantity of non-guaranteed matching drawn spots, the first quantity being at least one, the second quantity being at least one, wherein the display is after shuffling of an order of the guaranteed matching drawn spots for the play of the keno game and the non-guaranteed matching drawn spots for the play of the keno game, and wherein the display comprises no indications that any of the drawn spots for the play of the keno game are the guaranteed matching drawn spots for the play of the keno game, a display, by the display device, of an amount based on the drawn spots that match the player spots; and cause a display, by a display device, of an indication of an outcome of the play of the tournament.

18. The gaming system of claim 17, wherein for an additional one of the plays of the keno game for the play of the tournament, the plurality of instructions, when executed by the processor, cause the processor to cause: a display, by the display device, of the keno card comprising the plurality of spots, a display, by the display device, of indications of each of a plurality of player spots selected from the plurality of spots of the keno card, a display, by the display device, of indications of each of a plurality of drawn spots selected from the plurality of spots of the keno card, wherein all of the quantity of drawn spots comprise non-guaranteed matching drawn spots, and a display, by the display device, of an amount based on the drawn spots that match the player spots.

19. The gaming system of claim 17, wherein the plurality of instructions, when executed by the processor, cause the processor to, for the play of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for the play of the keno game to be at least equal to a quantity of drawn spots needed for an amount greater than zero to be won for the play of the keno game.

20. The gaming system of claim 17, wherein the plurality of instructions, when executed by the processor, cause the processor to, for the play of the keno game for the play of the tournament, cause the first quantity of guaranteed matching drawn spots for the play of the keno game to be less than a quantity of drawn spots needed for an amount greater than zero to be won for the play of the keno game.
